

Admissibility at Execution Time

A Verifiable Governance Model for AI Systems Based on Cryptographic Receipts and Human Oversight

ABSTRACT

Current AI governance frameworks emphasize validation, auditability, and post-hoc accountability. However, they do not guarantee that system actions are admissible at the moment of execution. This gap allows systems to produce outcomes without provable authorization under the governing conditions active at runtime.

This paper introduces a governance model centered on admissibility at execution time, where every action must be justified by verifiable authority before it becomes real. The model is implemented through a set of non-overridable invariants, a Proof of Human Oversight (PHO) mechanism, and a cryptographic forensic ledger that records execution receipts as immutable evidence.

We present an empirical methodological hardening event triggered by external adversarial review, in which identified epistemic and procedural fissures were formally integrated into the system through deterministic updates. The transformation is captured and sealed as a verifiable receipt, establishing a reproducible chain of governance evolution.

Keywords: AI governance, runtime enforcement, human oversight, cryptographic evidence, admissibility, EU AI Act

1. Introduction

The rapid deployment of AI systems in decision-critical environments has exposed a fundamental limitation in current governance approaches. Existing frameworks focus on whether systems are well-designed, tested, or auditable, but do not address a more immediate and decisive question:

Was the system allowed to act at the moment it acted?

This distinction separates two fundamentally different concerns: (1) whether a system can generate valid outputs, and (2) whether it is authorized to execute those outputs under current conditions.

Most governance architectures implicitly collapse these layers. Validation ensures that a system can behave correctly, while auditability ensures that its behavior can be reconstructed after the fact. Neither guarantees that an action was admissible at the exact moment of execution.

This paper proposes a different approach: **admissibility as a first-class runtime property**.

In this model, an action is not considered valid because it can be explained or audited later. It is valid only if it can be proven, at the moment of execution, that: (1) the action is authorized under the governing rules, (2) the authorization is non-overridable, and (3) the authorization is recorded as immutable evidence.

The central claim of this paper is that **governance must be resolved at execution time, not inferred afterward.**

2. Conceptual Distinction: Validation, Ownership, and Admissibility

2.1 Validation: Capability Without Authority

Validation determines whether a system behaves as intended under predefined conditions. It is inherently pre-execution or test-bound. A system can be fully validated and still produce an inadmissible action if runtime conditions invalidate its authority to act.

2.2 Ownership: Responsibility Without Constraint

Ownership addresses the attribution of consequences. It becomes relevant after an action has been executed. Ownership is therefore post-execution. It assigns responsibility but does not enforce constraints at the moment of decision.

2.3 Admissibility: Authority at Execution Time

The property that an action is authorized under the governing conditions at the exact moment of execution.

Unlike validation and ownership, admissibility is: runtime-bound (evaluated at execution time), state-dependent (based on current conditions, not prior tests), and binary (an action is either admissible or not).

2.4 Requirement: Admissibility as a First-Class Property

To resolve this gap, admissibility must be elevated to a first-class property of system design: (1) Every execution must be preceded by an admissibility decision, (2) This decision must be non-overridable under defined invariants, (3) The decision must produce verifiable evidence at the moment of execution.

3. The Governance Model

3.1 Architectural Overview

The system is structured as a layered execution pipeline:

USER → INTENT → COUNSEL → DOMAIN → LEDGER → VERIFY

Each stage refines a proposed action, but execution is not finalized until it is committed to the ledger. **An action becomes real only if it produces a valid receipt at commit time.**

3.2 Constitutional Invariants

At the core of the model is a set of constitutional invariants — system-level constraints that cannot be bypassed, overridden, or deferred.

ID	Name	Function
I6	Conflict Exposure	The system must surface internal contradictions. Suppression is not permitted.
I9	Prohibition of Autonomy Escalation	The system cannot elevate its own authority. Higher authority requires explicit human approval.
	Permanence of	All execution-relevant events must produce immutable, hash-linked

I11	Cryptographic Evidence	evidence.
I14	Explicit Failure Principle	When conditions for valid execution are not met, the system must fail explicitly.

3.3 Proof of Human Oversight (PHO)

For actions requiring human authority, the model introduces mandatory Proof of Human Oversight (PHO). An action requiring human approval must satisfy: (1) Identity — a valid human actor is identified, (2) Intent — the action is explicitly approved, (3) Presence — the approval occurs within the active execution context.

PHO ensures that human authority is not inferred or assumed, but proven at the moment of decision.

3.4 Forensic Ledger as Execution Gate

The forensic ledger serves three simultaneous roles: commit layer (finalizes execution), evidence layer (records immutable proof), and verification anchor (enables independent validation).

An action that does not produce a valid receipt is considered non-existent from a system perspective.

3.5 Non-Overridability

A critical property is that admissibility enforcement is non-overridable: Invariants cannot be disabled at runtime, PHO cannot be bypassed for actions that require it, and Ledger commitment cannot be skipped or deferred.

System integrity is defined not by compliance with policies, but by inability to violate admissibility conditions.

4. Empirical Case: Methodological Hardening Event

4.1 Context and Trigger

On April 25, 2026, the system underwent an external adversarial review conducted in a structured session. The review was explicitly critical and aimed at identifying epistemic and methodological weaknesses.

4.2 Identified Fissures

Fissure	Description
Authorship Circularity (W8)	Classification logic risked reinforcing its own assumptions without sufficient external grounding.
Closed-Loop Governance (W9)	Governance mechanisms were internally consistent but insufficiently open to external challenge.
Irrecoverability Gap	Absence of formal criterion for excluding irrecoverable cases weakened statistical integrity.

4.3 Deterministic Integration of Corrections

Change	Description
W8	Explicit acknowledgment and structural mitigation of circular reasoning risks
W9	Repositioning to incorporate external critique as valid input condition
E5	Formal exclusion rule with 10% quantitative threshold
§4.4	Statistical power recalculation (κ , PABAK, confidence intervals)
§4.6	Transparent procedure for irrecoverability criterion
§5.4	Vault reproducibility commitment
§6	Lexicographic precedence for ambiguous outcomes

Document versions updated: Eligibility Criteria v1.2 → v1.3, Classification Protocol v1.1 → v1.2

4.4 Receipt as Verifiable Evidence

✓ SEALED IN FORENSIC LEDGER

Field	Value
Receipt ID	PoE-METHODOLOGY-HARDENING-20260425
Timestamp	April 25, 2026 (UTC)
Governance Level	HIGH

The receipt is publicly accessible and independently verifiable.

4.5 Properties of the Event

Property	Description
Causality	Trigger explicitly linked to resulting changes
Determinism	All updates formally specified and version-controlled

Verifiability	Entire transformation anchored in accessible receipt
Non-Bypassability	Changes required satisfying invariant constraints and PHO
Reproducibility	Explicit commitments to reconstructability

5. Analysis

5.1 From Compliance to Admissibility Enforcement

Key distinction:

- **Compliance asks:** Did the system follow the rules?
- **Admissibility asks:** Was the system allowed to act at that moment?

This shifts governance from a descriptive layer to an operational gate.

5.2 Evidence as a First-Class Output

In the observed event: evidence is produced atomically with execution, evidence is cryptographically bound to the action, and absence of evidence implies non-existence of execution.

An action is only real if it is provable.

5.3 Governance Under Adversarial Pressure

The system did not adapt heuristically. It adapted through structured, admissible transformations.

Systems can evolve under adversarial conditions without losing structural integrity, provided that evolution itself is governed by admissibility constraints.

5.4 Determinism and Non-Bypassability

Rules that cannot be violated are more relevant than rules that can be audited.

5.5 Limitations

Limitation	Description
Infrastructure centralization	Verification depends on limited endpoints
Early-stage empirical base	Analysis based on single documented event
Operational dependency	Availability affects practical auditability

6. Implications and Conclusion

6.1 Implications for AI Governance Frameworks

Systems must prove, at the moment of execution, that they are allowed to act.

For regulatory contexts requiring human oversight (e.g., EU AI Act Article 14): human oversight cannot be treated as procedural requirement — it must be implemented as verifiable condition of execution.

6.2 Implications for System Design

Requirement	Description
Execution gating	Actions contingent on runtime authorization
Non-overridable constraints	Invariants that cannot be bypassed
Atomic evidence generation	Cryptographic proof at completion
Deterministic failure modes	Explicit failure when conditions not met

6.3 Future Work

Distributed verification (reducing centralized dependency), broader empirical validation (multiple domains and events), formal specification (machine-verifiable definitions), and standards integration (alignment with existing frameworks).

6.4 Conclusion

This paper introduced a governance model in which admissibility is enforced at execution time through invariants, human oversight, and cryptographic evidence.

An action is only valid if it can be proven, at the moment of execution, that it was authorized.

This shifts governance from retrospective assessment to deterministic control of what is allowed to become real.

In high-stakes environments, where consequences are immediate and irreversible, this distinction defines whether systems can be trusted not only to perform, but to act within provable bounds of authority.

Verification Appendix

The empirical case can be independently verified:

Receipt ID: PoE-METHODOLOGY-HARDENING-20260425

Public Verification URL:

<https://windi-domain.com/verify-public/?id=PoE-METHODOLOGY-HARDENING-20260425>

Document Hash Chain:

Document	Version	SHA-256
Eligibility Criteria	v1.2 → v1.3	57a84cc0... → 51acf3cd...
Classification Protocol	v1.1 → v1.2	2c704e66... → e7755c03...